

Large-Scale Product Review Sentiment & Fraud Detection

Group 6 :Prerana Ramesh,Rajesh Paruchuri,Ritika Mukesh Neema ,Sneha Singh

Abstract — Problem Definition

This project aims to design and implement a distributed big data system that analyzes large-scale product review data to perform sentiment classification and detect potentially fraudulent reviews. Using a publicly available Amazon product review dataset, we will process tens of gigabytes of review text using Apache Spark for large-scale cleaning, feature engineering, and aggregation. We will train machine learning models for sentiment prediction and anomaly/fraud detection. The system will include batch processing for historical training and a streaming component for real-time review scoring. The trained model will be deployed using FastAPI, with a lightweight web interface for predictions and visualization. MLflow will be used for experiment tracking, model versioning, and reproducibility.

Dataset Links and Short Description

Amazon Customer Reviews Dataset (AWS Public Dataset)

<https://registry.opendata.aws/amazon-reviews/>

This dataset contains millions of product reviews across multiple categories, including review text, ratings, timestamps, and reviewer information. The full dataset exceeds 10GB after decompression and requires distributed processing for analysis and model training.

Reference Links

Apache Spark: <https://spark.apache.org/>

MLflow: <https://mlflow.org/>

FastAPI: <https://fastapi.tiangolo.com/>

AWS Public Datasets: <https://registry.opendata.aws/>